

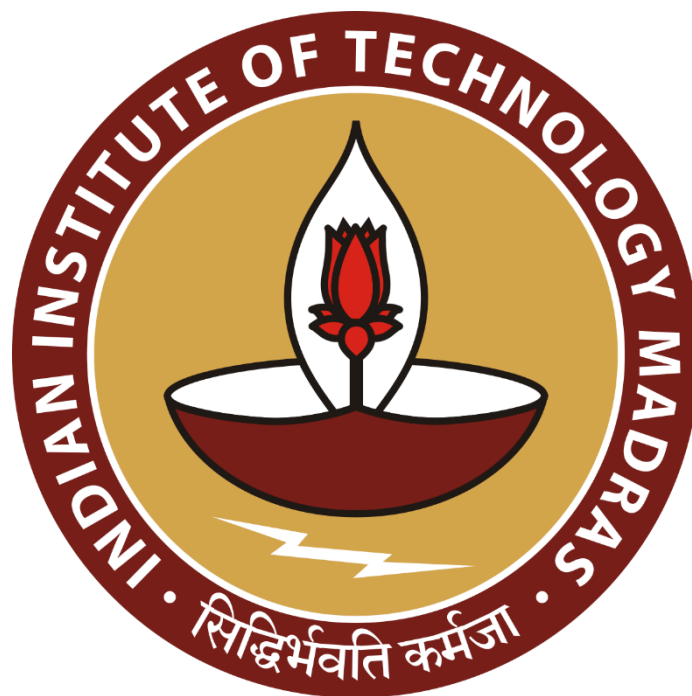
“Optimizing Inventory Management and Profitability for Sudha Milk Parlor”

A Mid-Term Report for the BDM Capstone Project

Submitted by

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Contents

1	Executive Summary	3
2	Proof of Originality of Data	3
3	Metadata	4
4	Descriptive Statistics	7
5	Detailed Explanation of Analysis Process	8
6	Results and Findings	9

1 Executive Summary

The mid-term report highlights the progress made in the project titled, “Optimizing Inventory Management and Profitability for Sudha Milk Parlor”.

The primary objective of this project is to come up with a solution that can address two major problems faced by Sudha Milk Parlor: inventory mismanagement (inefficient utilisation of the limited inventory) and product wastage (due to the short shelf-life nature of dairy products). As the first step of the project, sales data of various milk products sold in the shop were collected from the shop owner. As the collected data was handwritten, it was then digitized and converted into an Excel document to make further analysis of data simpler and more efficient.

The initial analysis performed on the data indicated that certain milk products are sold in significantly lesser quantities than others, thus leading to unnecessary financial investment in slow-moving products and inefficient inventory utilization.

The key outcome of this phase of the project is to analyze the sales performance of different milk products sold in the shop and distinguish best-selling products from underperforming products. This distinction is crucial as it forms the foundation for the final phase of the project.

2 Proof of Originality of Data

The data used in this project was provided by the owner of Sudha Milk Parlor, Mr. Vasant Kumar Singh, in handwritten form (click [here](#) to view the handwritten data logs). The dataset includes information about their sales records covering a three-month period from December 2024 to February 2025. Based on a review comment regarding the initial dataset size, a request was made to the shop owner for additional data. The owner agreed and provided older records to enhance the analysis.

The dataset consists of the following:

- Sales Data: The daily quantities of different products sold, the daily revenue generated per product and the overall revenue generated each day.
- Purchase Data: The monthly procurement details (if any), including the quantities of products purchased and their corresponding overall procurement costs.

(NOTE: While the sales data is the primary focus of this project, purchase data was collected as a supplementary dataset to analyze product turnover and assess individual product profitability.)

In order to ensure the originality of the data that was being collected, the following measures were followed:

- The data was recorded from photographs of Sudha Milk Parlor’s handwritten data logs.
- The data was manually digitized as an Excel document for analysis
- The final dataset was cross-verified by the shop owner to ensure its accuracy.

Additionally, the following supporting documents are submitted along with this mid-term report as proof of data authenticity:

- Photographs of the original handwritten data logs provided by the shop owner.
- An authorization letter from the shop owner confirming that the data used in this project is authentic and legitimate.
- A video recording of the discussion with the shop owner, covering details about the shop’s operations and challenges.
- A few photographs of the shop.

All the above mentioned documents are available at this [link](#).

3 Metadata

Source

The data was collected from the owner of Sudha Milk Parlor, Mr. Vasant Kumar Singh. The data conveys information about the sales and purchase records of the shop over a period of 3 months. The digitized dataset, converted to an Excel document can be found [here](#).

Format

The data is stored in the form of a table, with each record representing daily sales information.

Columns and Descriptions

The dataset comprising of the sales records is structured with the following columns:

Column Name	Description	Data Type
Date	The transaction date of sales (DD-MM-YYYY format)	Date
SM 1_Quantity	The number of Shakti Milk 1L units sold on that date	Numerical
S M 1_Revenue	The total revenue generated from Shakti Milk 1L sales	Currency (INR)

GOLD 500_Quantity	The number of Sudha Gold Milk 500mL units sold on that date	Numerical
GOLD 500_Revenue	The total revenue generated from Sudha Gold Milk 500mL sales	Currency (INR)
GOLD 1_Quantity	The number of Sudha Gold Milk 1L units sold on that date	Numerical
GOLD 1_Revenue	The total revenue generated from Sudha Gold Milk 1L sales	Currency (INR)
S M 500_Quantity	The number of Shakti Milk 500mL units sold on that date	Numerical
S M 500_Revenue	The total revenue generated from Shakti Milk 500mL sales	Currency (INR)
COW 500_Quantity	The number of Cow Milk 500mL units sold on that date	Numerical
COW 500_Revenue	The total revenue generated from Cow Milk 500mL sales	Currency (INR)
COW 1_Quantity	The number of Cow Milk 1L units sold on that date	Numerical
COW 1_Revenue	The total revenue generated from Cow Milk 1L sales	Currency (INR)
Total Revenue	The total revenue generated on that date	Currency (INR)

The dataset comprising of the monthly records of the purchase records is structured with the following columns:

Column Name	Description	Data Type
Month and Year	The month and year of purchase records	Date
S M 1_Purchase Quantity	The total quantity of Shakti	Numerical

	Milk 1L purchased in that month	
S M 1_Purchase Cost	The total cost incurred for purchasing Shakti Milk 1L in that month	Currency (INR)
GOLD 500_Purchase Quantity	The total quantity of Sudha Gold Milk 500mL purchased in that month	Numerical
GOLD 500_Purchase Cost	The total cost incurred for purchasing Sudha Gold Milk 500mL in that month	Currency (INR)
GOLD 1_Purchase Quantity	The total quantity of Sudha Gold Milk 1L purchased in that month	Numerical
GOLD 1_Purchase Cost	The total cost incurred for purchasing Sudha Gold Milk 1L in that month	Currency (INR)
S M 500_Purchase Quantity	The total quantity of Shakti Milk 500mL purchased in that month	Numerical
S M 500_Purchase Cost	The total cost incurred for purchasing Shakti Milk 500mL in that month	Currency (INR)
COW 500_Purchase Quantity	The total quantity of Cow Milk 500mL purchased in that month	Numerical
COW 500_Purchase Cost	The total cost incurred for purchasing Cow Milk 500mL in that month	Currency (INR)
COW 1_Purchase Quantity	The total quantity of Cow Milk 1L purchased in that month	Numerical
COW 1_Purchase Cost	The total cost incurred for purchasing Cow Milk 1L in that month	Currency (INR)

Timeframe

The data was collected over a 3-month period (from December 2024 to February 2025), giving a decent idea about the general sales trend of the shop.

Scope of Analysis

The collected dataset can be used to extract the following key information about the business:

- Identification of demand trends for each product
- Profitability analysis of individual products
- Scope for inventory optimization by comparing sales and purchase data of individual product

4 Descriptive Statistics

The dataset comprising the sales records consists of 90 records representing the daily sales transactions for a period of 3 months - from December 2024 to February 2025. It includes details on the quantity of individual milk products sold daily, their corresponding revenue and the overall daily revenue generated.

Overview of Dataset

- Total Records: 90
- Timeframe: December 2024 - February 2025
- Milk Products Studied:
 - (a) Sudha Gold Milk - 500 millilitre and 1 litre (6% fat)
 - (b) Shakti Milk - 500 millilitre and 1 litre (4.5% fat)
 - (c) Cow Milk - 500 millilitre and 1 litre (3% fat)

Key Statistics

- Total Revenue Generated: ₹1,64,187
- Average Daily Revenue: ₹1,824
- Median Daily Revenue: ₹1,878
- Highest Single-Day Revenue: ₹2,701 (6th January, 2025)
- Lowest Single-Day Revenue: ₹664 (25th February, 2025)

Product-Wise Revenue Contribution

- Highest Contribution: Cow Milk (1 Litre): 34.45% of total revenue
- Lowest Contribution: Cow Milk (500 millilitre): 6.16% of total revenue

Sales Quantity Analysis

Total Quantities Sold: 4632

Best-Selling Product: Sudha Gold Milk (500 millilitre) - 99.88% of purchased stock sold

Least-Selling Product: Sudha Gold Milk (1 litre) - 94.44% of purchased stock sold

Data Visualization

To better understand the sales trends and inventory challenges, the following analyses were conducted:

- Sales Performance Analysis
 - To identify the best-selling and slow-moving products
- Profitability Analysis
 - To evaluate the profit contributions from individual products
- Purchase vs Sales Quantity Comparison
 - To highlight inefficiencies in inventory utilization

5 Detailed Explanation of Analysis Process

The analysis in this project is a multi-step approach to identify slow-moving products, evaluate profitability of individual products and assess the efficiency of inventory utilization.

Data Formatting and Cleaning

The handwritten data collected from the shop owner was first converted into an Excel document. The dataset was structured into tabular format with the relevant column names (as described in Section 3).

- Inconsistencies in product names (e.g., “SM1” and “Shakti 1”) were identified and standardized (15 entries, about 16.67% of the dataset).
- Unclear numerical entries (e.g., “3” misread as “8”) were identified and verified with the shop owner for clarity (10 entries, about 11.11% of the dataset).

Exploratory Data Analysis (EDA)

Once the dataset was formatted and cleaned, descriptive statistics was calculated to get a basic idea about the product-wise performance in the overall profitability of the shop. Then, the following parameters were deduced from the dataset:

- Product-wise revenue contribution was analyzed to identify best-selling and slow-moving products.
- Sales conversion rates (sold quantity vs. purchased quantity) were computed to assess inventory turnover efficiency.
- Profit margins for individual products were derived by comparing selling prices with estimated purchase costs.

Tools Used

- **Python:** Data Cleaning, Data Visualization, Analysis
- **Microsoft Excel:** Data Structuring, Data Formatting

6 Results and Findings

Based on the analysis performed on the data, the following results were obtained:

(i) Product-Wise Revenue Contribution

- Cow Milk (1L) was the highest contributor to the overall revenue, accounting for about 34.5% of the total revenue.
- Sudha Gold Milk (500mL) was the second-highest contributor, accounting for about 26.3% of the total revenue, followed by Shakti Milk (1L) at 19.4%.
- Together, these three best-selling products accounted for 80.2% of the shop's total revenue, making them the key revenue drivers.
- On the other hand, Cow Milk (500mL), Shakti Milk (500mL) and Sudha Gold Milk (1L) were the least-selling products, contributing only 19.8% of the total revenue.

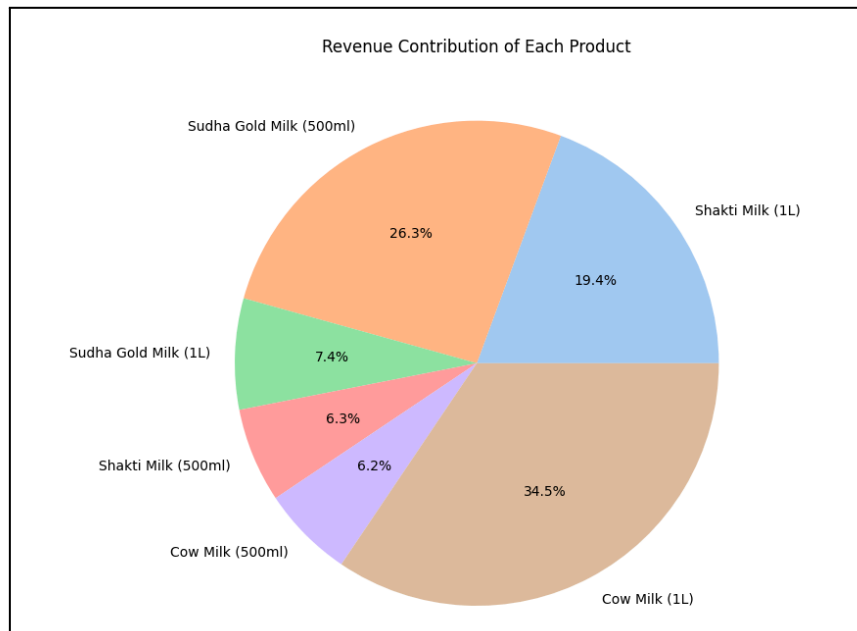


Figure 1: Revenue Contribution of Each Product

(ii) Sales Conversions Rates

- Shakti Milk (1L), Sudha Gold Milk (500mL), and Cow Milk (1L) had sales conversion rates above 99.8%, implying that almost all purchased stock was sold (with only about 5 unsold units over 3 months). Hence, these products can be considered as fast-moving products.
- In contrast, Sudha Gold Milk (1L), Shakti Milk (500 mL) and Cow Milk (500mL) were all slow-moving products, with 41 unsold units over the 3 months period. It is clear that these products not only reduce profitability but also occupy valuable inventory space, leading to potential losses to the shop.

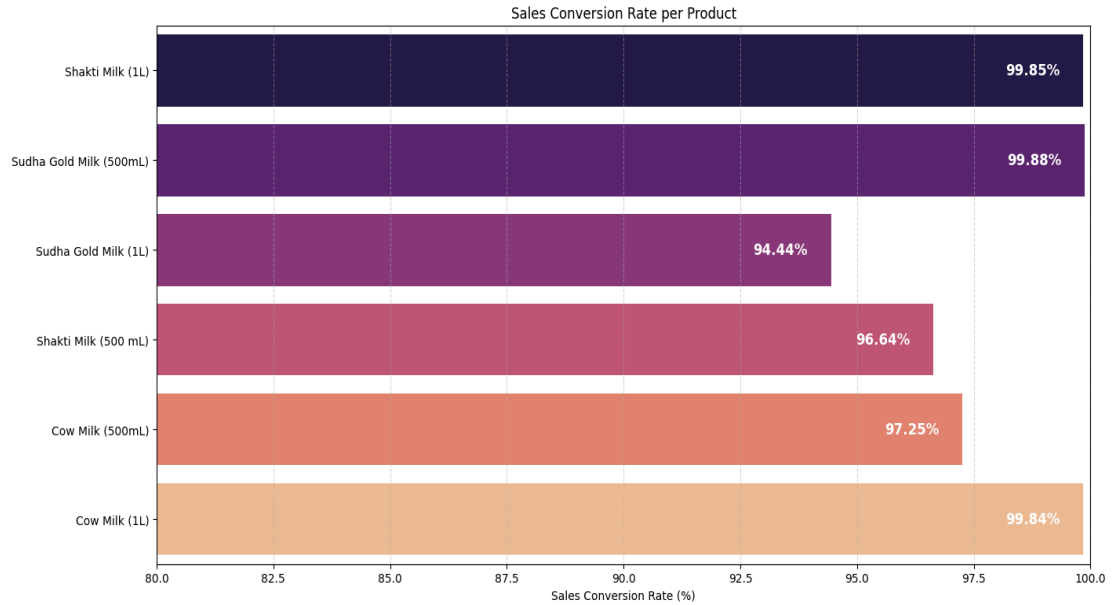


Figure 2: Sales Conversion Rate per Product

(iii) Profit Margins for Individual Products

- Sudha Gold Milk (500mL) generated the highest profit for the shop, followed by Cow Milk (1L).
- Sudha Gold Milk (1L) incurred a net loss, while Shakti Milk (500mL) and Cow Milk (500mL) had the lowest profit margins, making them the least profitable products.

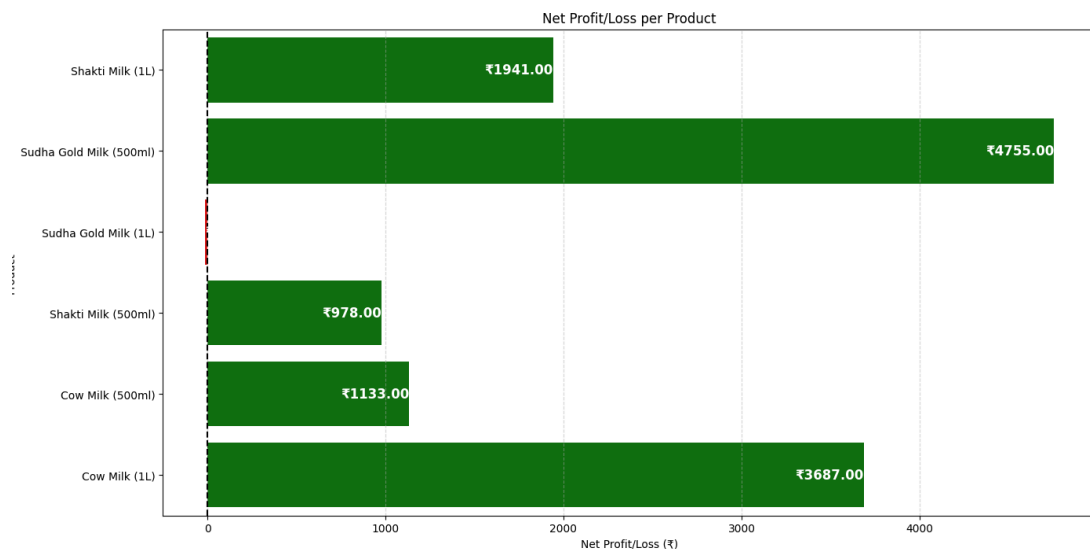


Figure 3: Net Profit/Loss per Product

Conclusions

Best-Selling and High-Profit Products

- Shakti Milk (1L), Sudha Gold Milk (500mL) and Cow Milk (1L) are the best-selling products in the shop, accounting for a larger portion of the shop's profit.

- These products need to be purchased in higher numbers to further maximize profit and revenue.

Slow-Moving and Low-Profit Products

- Sudha Gold Milk (1L), Shakti Milk (500mL) and Cow Milk (500mL) are the slow-moving products in the shop which not only hinder the profitability of the shop, but also occupy valuable inventory space.
- These products need to be purchased in lesser quantities and a dynamic pricing strategy must be considered to be implemented for these products.